

TECHNICAL PROPOSAL COVER SHEET

NAME OF COMPANY: George Washington University Medical Faculty Associates

ADDRESS: 2150 Pennsylvania Ave 5-416 Washington, DC 20037

WEBSITE: <https://changlab.smhs.gwu.edu/>

TYPE OF BUSINESS: Non-profit

BUSINESS SIZE: Large, independent physician entity based in Washington, D.C. employs over 2,150 individuals, including more than 750 providers across 51 medical specialties.

DUNS NUMBER: DUNS: 841694974

TITLE OF PROPOSAL: Identification of early symptomatic biomarkers and correlates of disease severity utilizing samples collected during the current dengue outbreak in Colombia and Thailand

PROPOSED AMOUNT: \$110,000 in Year 1 (FY2025) and \$55,000 in Year 2 (FY2026)

PERIOD OF PERFORMANCE: TBD

TECHNICAL POINT OF CONTACT: Aileen Chang, MD MSPH, Department of Medicine, chang@email.gwu.edu

BUSINESS POINT OF CONTACT: Michelle Van Fossen, Office of Clinical Research/MFA, mvanfossen@mfa.gwu.edu

1. Introduction/Background

This contract is for a MIDRP proposal MI240174 entitled Identification of early symptomatic biomarkers and correlates of disease severity utilizing samples collected during the current dengue outbreak in Colombia and Thailand.

This study aims to identify early biomarkers (within the first 96 hours of onset of disease) associated with severe disease manifestations for patients with dengue infection. The goal is to use these prognostic markers to stratify patients for treatment. Dengue fever is a nationwide, year-round disease in Thailand and Colombia with peak transmission occurring in the rainy seasons each year. Retrospective and prospective cohort human samples collected from dengue patients classified into dengue without warning signs (DEN-WS), dengue with warning signs (DEN+WS), and severe dengue (SD) will be acquired from Colombia and Thailand. Immunological markers, i.e. antibody neutralization (Neut), antibody-dependent enhancement (ADE), serum NS1 and anti-NS1 IgG/IgA, and acute phase proteomics (Olink) and transcriptomics (RNAseq) will be investigated. Biostatistics will be used to discover and evaluate markers that are associated with clinical endpoints.

The study will be carried out through a collaboration of NMRC with George Washington University. During the 2-year period of the project, the objectives are to obtain approximately a minimal of 200 cases (n=100 of DF-WS; n=100 of DF+WS and SD) for serum/plasma and

PBMCs. Since the rate of SD is commonly low, to mitigate the risk of low SD, GW will try to increase the study size to >200. All samples will be shipped to NMRC for the completion of the research objectives at NMRC.

This contract is between NMRC and GWU.

The Colombian site, led by Dr. Aileen Chang, an Associate Professor of Medicine and Microbiology, Immunology and Tropical Medicine at George Washington University (GWU). Dr. Chang has been working in partnership with Colombian institutions for the last decade and has extensive experience executing observational and interventional clinical trials in Barranquilla, Colombia. In particular, she is currently working on a clinical trial in hospitalized patients in Colombia and has experience recruiting severe dengue cases.

2. Technical Requirements

GWU will provide samples which have been already collected and will continue conducting the prospective patient enrollment. GWU will collect serial samples at Day 0 (With fever in the last 7 days), Day 2, and convalescent (45-90 days after infection). Demographic data and clinical characteristics including warning signs, symptoms of severe dengue and results of diagnostic laboratories and clinically significant laboratory abnormalities and infection outcomes.

Requirement:

- 2.1 GWU will send samples collected in 2024-2025 by March/ April 2025 together with the related clinical and laboratory records. The expected number of subjects is ≥ 30 .
- 2.2 GWU will send samples collected in 2025 by Sep/ Oct 2025 together with the related clinical and laboratory records. The expected number of subjects is ≥ 70 .
- 2.3 GWU will send sample collected during 2025-2026 by Jun/Jul 2026. The expected number of subjects is ≥ 100 .

Note: Funding for 2.1 and 2.2 has been received. Funding for 2.3 has not been received and will be conditional based on funding.

3. Deliverables/Schedules

- 3.1. Mar/Apr 2025: A shipment of samples described in 2.1.
- 3.2. Sep/Oct 2025: A shipment of samples described in 2.2.
- 3.3. Jul/Aug 2026: A shipment of samples described in 2.3.

Note: Deliverables 3.3 will be determined by funding as described in 2.

4. Applicable Documents

Quotation documents from the contract (attached below).

5. Monitoring the Progress

Emails, TEAM meetings, and phone calls will be the means for communication to exchange information on issues or progress during the contract execution.

6. Acceptance Criteria

Quality and quantity of the products specified on the quotation document.